~~* Discuss about the component used in next event time advance approach.~~

~~Ans: The components are:~~

~~1. Simulation clock~~

~~2. Event list~~

* What are the steps of simulation for single server queuing system? Describe a problem statement and intuitive explanation of single server queuing system. 2021 → 18-19 - 1(b)

Ans: Steps:

1. Define problem and objective
2. Set up simulation environment
3. Generate random input.
4. Run the event processing loop
5. Determine termination condition.
6. Collect and analyze result.
7. Validate and verify simulation
8. Document and report finding.

Problem Statement:

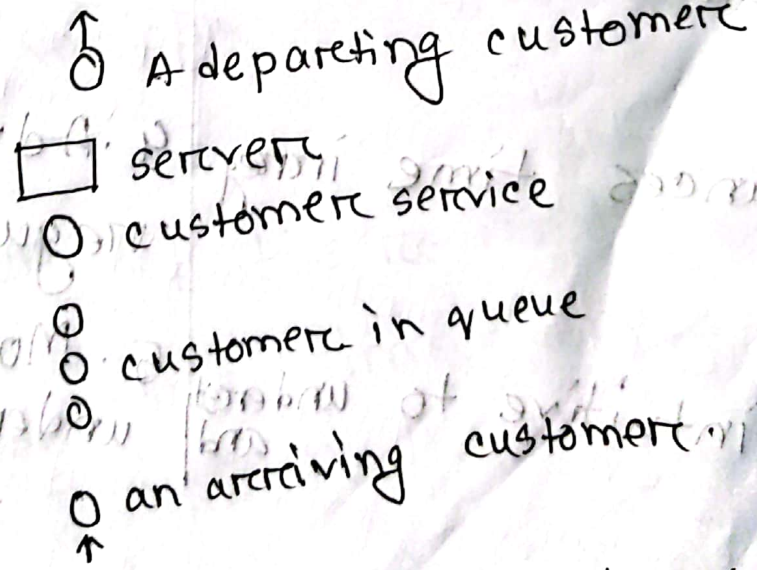


Figure: A single server queuing system.

Intuitive explanation: The intuitive explanation is ~~of~~ how to simulate a single server queuing system by showing how its simulation model would be represented inside the comput

*what are the difference between next event time advance approach and ~~f~~ fixed increment time advance approach? 17-18-2(a)

Next event time advance	fixed increment time adv
1. The simulation clock jumps directly to the time of next event.	1. The simulation clock jumps equal steps and time increment are fixe
2. It is complex	2. It is simple.
3. It is ~~sufficient~~ efficient with ~~when~~ irregular event.	3. It is inefficient with irregular event.
4. Advances time irregular.	4. Advances time in regular.
5. less intuitive to understand	5. More intuitive to understand.

<u>Mid</u>

Mahbub Sir – Chapter- 1 বই থেকে (D. S. Hira বই)

<u>Masud Sir</u> বই (Simulation Modeling and Analysis by Averill M. Law)

<u>Chapter- 1</u>

- Assumption: It usually take the form of mathematical or logical relationships, constitute a model that is used to try to gain some understanding of how the system behaves.

$$y = mx + c$$

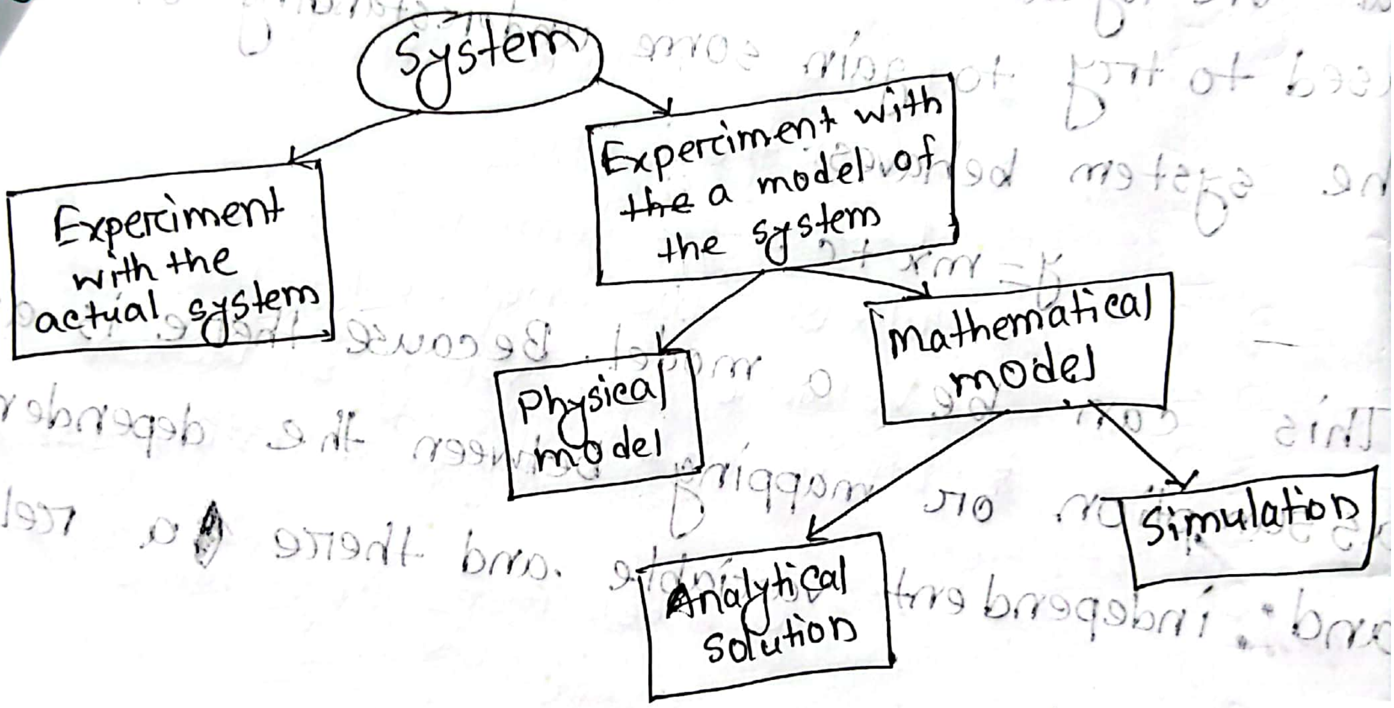
This can be a model. Because there is an assumption or mapping between the dependent and independent variable and there a relation between.

* Analytical solution: If the relationship ~~are~~ that compose the model are simple enough and it is possible to use mathematical method to obtain exact information on question of interest. It is called analytical solution

*Application of Simulation (system simulation by D.S Hira বই থেকে)

***System:** System is ~~defined as~~ a collection of components that act and interact together toward the accomplishment of some logical end.

***Ways to study a system:**



*Discrete Event Simulation → Example - 1.1.

* Time Advance Mechanism - " 1.2

In some discrete-event simulation models events are used for purposes that do not actually effect such a change.

*Experiment with Actual System vs Experiment with a model of the system :

Experiment with actual system	Experiment with a model of the system
1.	1.

*Analytical Solution vs simulation : 2018 – 1(b)

Analytical Solution	Simulation
1. If the relationship that compose the model are simple enough it is possible to use mathematical method to obtain exact information on question of interest.	1. Simulation is the representation of real life system by another system which depicts important characteristic of real life system and allows experiment on it.
2. In this, the model is simple.	2. In this the model is complex
3. It gives exact answer.	3. ~~Some~~ It doesn't give exact answer.
4. ~~Some~~ simple mathematical model is efficient for analytical solution.	4. ~~Some~~ ~~Simple~~ complex mathematical model is ~~insu~~ efficient for ~~sum~~ simulation

* Components of a Discrete-Event simulation model :

~~[2018-1(a)]~~ [2018-1(a)]

⇒ System state

⇒ Simulation clock

⇒ Event list

⇒ Statistical counters

⇒ Initialization routine

⇒ Timing routine

⇒ Event routine

⇒ Library routine.

⇒ Report generator

⇒ Main program.

*Discuss about the component used in next event time advance approach →17-18-1(a)

Mahbub Sir — Chapter-2 (D. S. Hira)

* Monte Carlo Method:

It is a method that uses simulated random numbers to estimate some functions or solve a problem.

* Solve the integral $I = \int_2^5 x^3 dx$ by monte carlo method. [USE: Random number 61 to 70 for X coordinate and Random Number 71 to 80 for Y coordinate, from table. [Previous mid-3]

Solution:

$$I = \int_2^5 x^3 dx$$

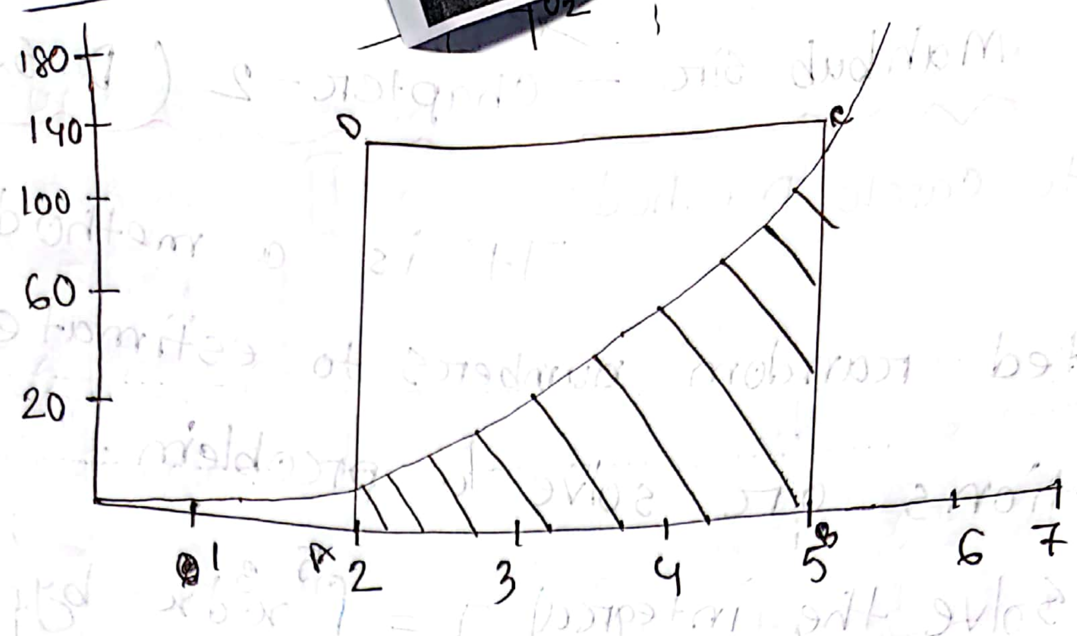
The exact value of the integral, $I = \left[\frac{x^4}{4}\right]_2^5$

$$= \frac{5^4}{4} - \frac{2^4}{4}$$

$$= 152.25$$

The function $f(x) = x^3$ is plotted in the below figure. The area under the curve between limits $x = 2$ and $x = 5$ shown shaded is the value of the integral.

fig - 8



Let the area be enclosed in a rectangle ABCD whose area is $140 \times 3 = 420$. Let N dots fall in the rectangle ABCD and let m dots fall in the shaded area. Then the ratio of the shaded area I to the area A of rectangle ABCD is m to N.

$$\frac{I}{A} = \frac{m}{N}$$

or, $$I = \frac{m}{N} A$$

Let us take random number between ~~20~~ 61 to 70 to represent x co-ordinate.

~~et~~ Now, to generate y coordinate, let's take random number between 71 to 80.

Table : Simulation of $I = \int_2^5 x^3 \, dx$

$y \le x^3$ $\to$ M+, M-

Random Number (R)	x ($0.1 \times R$)	Random Number (R)	y ($140 \times R$)	x^3	M	N
25	2.5	0.42	58.8 ~~42~~	15.625	0	1
15	1.5	0.10	14 ~~70~~	3.375	0	2
43	4.3	0.90	126	79.507	0	3
47	4.7	0.57	79.8	103.823	1	4
22	2.2	0.76	106.4	10.648	1	5
18	1.8	0.77	107.8	5.832	1	6
27	2.7	0.05	7	19.683	2	7
45	4.5	0.18	25.2	91.125	3	8
40	4.0	0.66	92.4	64	3	9
48	4.8	0.00	0	110.592	4	10

$$\therefore \frac{I}{A} = \frac{M}{N}$$

$$\Rightarrow I = \frac{M}{N} \times A$$

$$= \frac{4}{10} \times 420$$

$$= 168$$

* 8 A drunkard moved from a point to a direction ... [Previous mid-1]

Solution:

As drunkard takes step in four direction. Backward, forward, right, left. Probability are. 10%, 50%, 15%, 25%. Distance covered to backaward, forward, right, left are 30 cm, 80 cm, 60 c.m, 40 c.m.

Question অনুযায়ী serially বসাবে।

Direction	Probability	Random number
Forward (F)	0.5	0 – 49 ~~1 – 50~~ ~~61 – 80 c.m~~
Backward (B)	0.1	50 – 59 ~~1 – 30 c.m~~
Left (L)	0.25	60 – 84 ~~31 – 40 c.m.~~
Right (R)	0.15	86 – 99 41 – 60 c.m

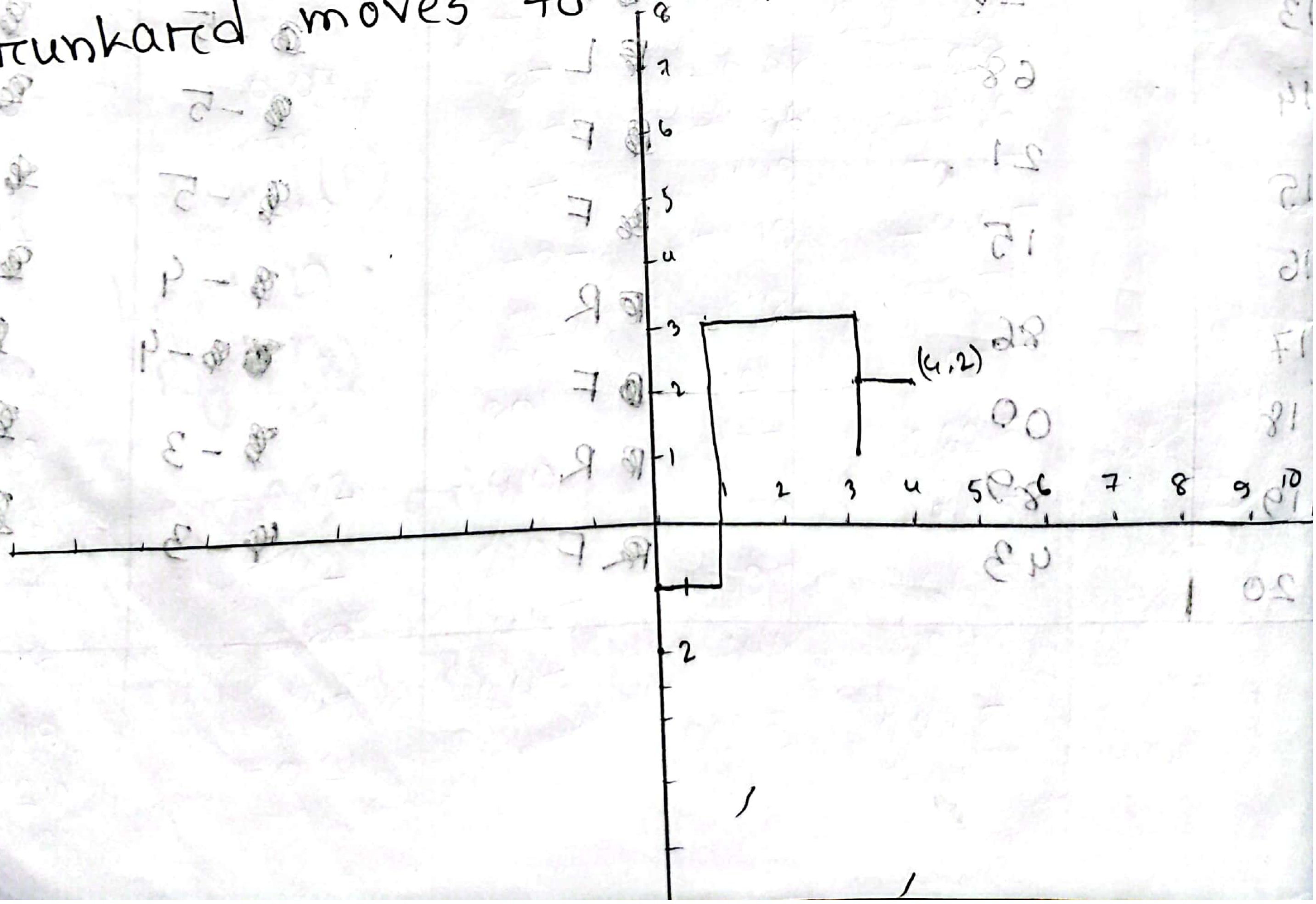
*Forward ($\overrightarrow{+1}$) গেলে y axis → 1 বাড়বে। Backward ($\overrightarrow{-1}$) গেলে y → 1 কমবে।

* Left গেলে x axis → −1 হবে। Right গেলে x-axis → +1 হবে।

Table: Random Walk Simulation

Step	Random Number	Direction	Position x	Position y
1	23	F	0	1
2	01	F	0	2
3	67	L	-1	2
4	56	B	-1	1
5	84	L	-2	1
6	95	R	-1	1
7	77	L	-2	1
8	76	L	-3	1
9	18	F	-3	2
10	82	L	-4	2
11	45	F	-4	3
12	55	B	-4	2
13	29	F	-4	3
14	68	L	-5	3
15	21	F	-5	4
16	15	F	-5	5
17	86	R	-4	5
18	00	F	-4	6
19	89	R	-3	6
20	43	F	-3	7

when the drunkard takes steps in forward direction, the value of y increased by 1. If drunkard takes steps in backward the value of y decreased by 1. If drunkard takes steps in left value of x decrease by 1 and for right value of x increased by 1. The simulation for 20 steps is given in table. In the first 20 steps, drunkard moves to a position (4,2)



* Previous mid - 2 - Bombing Problem.

The point of strike is assumed to be normally distributed around the aiming point with a standard deviation of 500m in x direction and 350 in y direction.

We know, A normal random variable X is given by

$$X = \mu + \sigma Z$$

where, μ is the true mean of the distribution,
σ is the standard deviation of x,
Z is the normal random number.

If x and y are the coordinates of a point.

then,
$$x = \mu_x + \sigma_x z_x$$
$$y = \mu_y + \sigma_y z_y$$

Taking coordinates of the center point of the area as 0, $\mu_x = \mu_y = 0$

The given value of standard deviation in X and Y directions are 500 and 350 m respectively.

$$x_1 = \boxed{500}\, z_x$$

$$y_1 = \boxed{350}\, z_y$$

marked area

Table : Simulation of Bombing Problem

Bomb Strike	RNN	x $(500 \times z_x)$	RNN	y $(350 \times z_y)$	Result
1	0.23	115	0.24	84	Hit ~~Miss~~
2	-1.16	-580	-0.02	-7	Miss
3	0.39	195	0.64	224	Hit
4	-1.90	-950	-1.04	-364	Miss
5	-0.78	-390	0.68	238	Hit
6	-0.02	-10	-0.47	-164.5	Hit
7	-0.04	-20	-0.75	-262.5	Hit
8	-0.66	-330	-0.44	-154	Hit
9	1.41	705	1.21	423.5	Miss
10	0.07	35	-0.08	-28	Hit
11	0.53	265	-1.56	-546	Miss
12	0.003	15	1.49	521.5	miss
13	-1.19	-595	-1.19	-416.5	Miss
14	0.11	55	-1.86	-651	Miss
15	0.16	80	1.21	423.5	miss
16	-0.82	-410	0.75	262.5	Hit
17	0.42	210	-1.50	-525	Miss
18	-0.89	-445	0.19	66.5	Hit
19	0.49	245	-1.44	504	Miss
20	-0.21	-105	0.65	227.5	Hit

Number of hits = 10

" " miss = 10

% Number of strikes on target = 50%

17-18, 20-19, 5(a), 2021, 3(b)

* Monte carlo method vs Stochastic simulation

Monte Carlo method	Stochastic simulation
1. It is used to obtain solution which are deterministic in nature.	1. It is used to obtain solution of problems which are stochastic in nature.
2. We can determine exact outcome at any time.	2. We can not determine exact outcome at any time
3. Applies to static model.	3. Applies to dynamic model
4. Example: area of an irregular figure, numerical integration, evaluation of π, trajectory computations etc.	4. Example: gambling game, bombing problem, reliability problem, inventory problem. etc

Chapter-4

*Mid Square Random Number Generator:

Steps are:

1. Starting with n digit number
2. Squaring it.
3. From the squared chop off the two low order digits and one ~~of~~ or two high order digits.

For 8 digit → Remove two lower and higher digits

For 7 digit → Remove one lower and two higher order digit

* What are the limitations of mid square method? Evaluate the performance of mid-square method if seeds are 6785, 1379, 1357.

2017-18 2(b)

Soln:

The limitations are:

1. The sequence of numbers is limited.
2. Convergence to zero after a few iterations.
3. Limited precision due to computer word size.
4. Predictable output, making it less random.
5. Unsuitable for large-scale or modern applications.

Let the seed number be 6785. When squared we get $= (6785)^2 = \underline{4}60362\underline{25}$.

After removing two low order digits and two high order digits, we get next random number 0362. Its square is $= (0362)^2 = 131044$.

~~After removing~~ one low ~~order and~~ one ~~two~~ high ~~orde digit, we get~~. This 6 digit number we can add 00 at the left and then removing 00 and 44 we get 1310. A sequence of 10 numbers with a seed of 6785 is:

0362 → 1310 → 7161 → 2799 → 8344

6223 → 7257 → 6640 → 0896 → 8028

Similarly for 1379 ∴ sequence of random number:

9016 → 2882 → 3059 → 3574 → 7734

8147 → 3736 → 9576 → 6997 → 9580

for 1357 :-

8419 → 7953 → 2502 → 2600 → 7600

7600 → 7600

* For Multiplicative Congruential Generator:

$$r_{i+1} = a r_i \bmod m$$

* For Additive Congruential Generator:

$$r_{i+1} = (r_i + b) \bmod m$$

~~* Perform the Kolmogorov - Smirnov~~

* Steps in Kolmogorov - Smirnov test:

⇒ Step-1: Rank the data from smallest to largest

$$R_{(1)} \leq R_{(2)} \leq \cdots R_{(N)}$$

⇒ Step-2: compute

$$D^+ = \max \left\{ \frac{i}{N} - R_{(i)} \right\}$$ ঋণাত্মক value নেয়া যাবে না

$$D^- = \max \left\{ R_{(i)} - \frac{i-1}{N} \right\}$$ " " " " "

⇒ Step-3: compute $D = \max(D^+, D^-)$

⇒ Step-4: Get D_α for the significance level α

⇒ Step-5: If $D \leq D_\alpha$ accept otherwise reject

* The following sequence of random numbers have been generated 0.037, 0.55, 0.71, 0.97, 0.65, 0.29, 0.84, 0.78, 0.23, 0.17. Use Kolmogorov Smirnov test with $\alpha = 0.05$ to determine. If these numbers are uniformity distributed over the interval 0 to 1. Note that $\alpha = 0.05$ and $N = 10$ the critical value is 0.410

$100-5 = 95$ 2017-18-14(b) 2015-16-3(a)

Soln:

i	1	2	3	4	5	6	7	8	9	10
R_i	0.037 ~~0.17~~	0.17	0.23	0.29	0.55	0.65	0.71	0.78	0.84	0.97
$\frac{i}{N}$	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$\frac{i}{N} - R_i$	0.063	0.03	0.07	0.11	—	—	—	0.02	0.06	0.03
$R_i - \frac{i-1}{N}$	0.037	0.07	0.03	—	0.15	0.15	0.11	0.08	0.04	0.07

$D = \max(0.11, 0.15) = 0.15$

critical value of D for $\alpha = 0.05$ is 0.410

since, $\underset{0.410}{D_\alpha} > \underset{0.15}{D}$, the given random numbers are uniform at 95% level of significance.

If critical value of D for $\alpha = 0.01$ is 0.368

then, $D_\alpha > D$, $0.368 > 0.15$, so given random numbers are uniform at 99% level of significance.

* The two digit random numbers generated by ~~multiplicative congruentio~~ a random number given. Determine Chi-square. Is it acceptable at 95% confidence level? 2016-17-4(c)

Soln: $\alpha = 0.05$ — 2018-19-6(b)

বইয়ের exercise -10

The given 100 random numbers can be divided into 10 classes as given below:

Class	Count	Frequency	Diff	Diff^2
$0 < r \leq 10$	**********	10	0	0
$10 < r \leq 20$	************	12	2	4
$20 < r \leq 30$	************	12	2	4
$30 < r \leq 40$		7	3	9
$40 < r \leq 50$		5	5	25
$50 < r \leq 60$		10	0	0
$60 < r \leq 70$		11	1	1
$70 < r \leq 80$		10	0	0
$80 < r \leq 90$		16	6	36
$90 < r \leq 100$		7	3	9
				= 88

$\therefore$ Chi-square $= \frac{88}{10} = 8.8$

For degree (10-1) = 9 of freedom, at 95% confidence level, the acceptable value of chi-square is 16.9 which is more than 8.8. Hence given random number is acceptable.

2015-16 - 3(b)

* Generate two digit 40 random numbers between 0 to 50 by multiplicative congruential method. Apply Chi-Square test to verify the randomness. Where, the accepted value of chi-square for four degree of freedom at 95% confidence level is 9.448.

2018-10- 6(b·ii)

***Serial Auto correlation:** When in a sequence of number, the numbers are not independent and one number decides the next then the defect is called ~~A~~ serial autocorrelation

* Test the autocorrelation of the given numbers ~~by~~ বই এর ~~ex~~ ch-4 eex-10 এর মত. employing the chi-squared test with 99% confidence level.

Soln: For this test the overlapping pairs of random numbers are to be taken. Taking three classes; 0 to 33 ; 34 to 67; 68 to 100 for each random number in the pair, the number of pairs in each class is obtained as follows. Since there are 9 classes, the expectation value is 10

R_1	R_2	Count	Frequency	Diff	$Diff^2$
≤ 33 &	≤ 33	* * * * * * * * * * * * * * * *	~~16~~ 16		
≤ 67 &	≤ 33	* * * * * *	~~20~~ 6		
≤ 100 &	≤ 33	* * * * * * * * * * * * *	~~32~~ 13		
≤ 33	≤ 67	* * * * * *	6		
≤ 67	≤ 67	* * * * * * * * * * * *	12		
≤ 100	≤ 67	* * * * * * * * * * *	11		
≤ 33	≤ 100	* * * * * * * * * * * * * *	14		
≤ 67	≤ 100	* * * * * * * * *	9		
≤ 100	≤ 100	* * * * * * * * * * * * * * *	15		

R_1, R_2 হলো ছকে যে sequence এ
value দেয়া থাকবে সেখানে first value (R_1)
থেকে second value (R_2) compare করবে
R_1 & R_2 দুটো শর্তই মানতে হবে। যেমন
values এ first value $R_1 = 21$ and $R_2 = 81$
এটা first শর্ত পূরণ করে না তাই accepted
না আবার 81 এর সাথে 92 compare এভাবে
count করতে হবে।

* A sequence of 10,000 five digit random number has been generated

2017-18 – 3(c), 16-17 – 4(b)

Soln:

The calculation of χ^2 is given below:

Combination Distribution	Observed value (O_i)	Expected value (E_i)	$\frac{(O_i - E_i)^2}{E_i}$
Five different digit	3075	3024	0.8601
Pairs	4935	5040	2.1875
Two-pairs	1135	1080	1.8750
Three of kinds	695	720	0.868
Full houses	105	90	2.5
Four of kinds	54	45	1.8
Five of kinds	1	1	0
	10,000	10,000	10.0907

The appropriate degree of freedom in this case are 6, one less than the number of combinations.

The critical value of x^2 for six degree of freedom at $\alpha = 0.01$ is 16.8. The value of χ^2 obtained from calculation is 10.097 which is less than the critical value. Hence; Hypothesis of independence can not be rejected on the basis of Poker test.

* 4 ~~dif~~ digit এর ক্ষেত্রে:

	E_i
⇒ four different digit	5040
⇒ one pair in the digit	4320
⇒ Two pairs	540
⇒ Three ~~di~~ of kinds	90
⇒ Four of kinds.	10

⇒ three digit num:

⇒ three different "	720
⇒ one pair "	270
⇒ Three of a kind	10

* Use multiplicative congruential method to generate a sequence of 10 three digit random number, Let, $r_0 = 117$, $a = 3$, $m = 1000$

Soln:

We know,

$$r_{i+1} = a r_i \text{ modulo } m$$

$1000 \overline{)351(} 0$
$\underline{351}$

here, $a = 3$, $r_0 = 117$, $m = 1000$

351

$\therefore r_1 = ar_1 \text{ modulo } m = 3 \times 117 \bmod 1000 = 351 \bmod 1000 = 351$

$r_2 = 351 \times 3 \bmod 1000 = 1053 \bmod 1000 = 053$

$r_3 = 3 \times 53 \bmod 1000 = 159 \bmod 1000 = 159$

$r_4 = 3 \times 159 \bmod 1000 = 477 \bmod 1000 = 477$

$r_5 = 3 \times 477 \bmod 1000 = 1431 \bmod 1000 = 431$

$r_6 = 3 \times 431 \bmod 1000 = 1293 \bmod 1000 = 293$

$r_7 = 3 \times 293 \bmod 1000 = 879 \bmod 1000 = 879$

$r_8 = 3 \times 879$ " " $= 2637 \bmod 1000 = 637$

$r_9 = 3 \times 637$ " " $= 1911 \bmod 1000 = 911$

$\cancel{r_{10} = 3 \times 911 \text{ " " } = 2733 \text{ " " } = 733}$

The sequence numbers of 10 numbers are,
117, 351, 53, 159, 477, 431, 293, 879, 637, 911

Masud Sir

Chapter - 7

*Random variates: It is a outcome of a random variable. ~~Random variat~~

*Random variable/number: Random variates generated ~~for~~ from the U(0,1) distribution will be called random number.

*A good arithmetic random number generator should posses several properties:

1. The number produced should be distributed uniformly on [0,1], should not exhibit any correlation with each other.
2. The generator should be fast and avoid lots of storage.
3. Reproduction of random numbers are for two reasons: i) debugging or verification of computer program easily.
ii) identical random number in simulating system to obtain more precise comparison.

4. There should be provision in the generator for easily producing separate ~~stem~~ streams

5. We should like the generator to be portable

*Linear Congruential Generator (LCG):

A sequence of integer $z_1, z_2, \ldots$ is defined by the recursive formula,

$$z_i = (a z_{i-1} + c) \pmod m$$

$$U_i = \frac{z_i}{m}$$

where, m is the modulus, a the multiplier, c is the increment, z_0 = the seed or starting value,

should satisfy: $0 < m$

$$a < m$$

$$c < m$$

$$z_0 < m$$

Example 7.2 Consider the LCG defined by $m=16$, $a=5$, $c=3$ and $Z_0=7$. table below gives Z_i and U_i for $i=1,2,\ldots,19$.

16 | 38 |

16 | 8 | 0

i	$Z_i=(5Z_{i-1}+3) \bmod 16$	$U_i=\frac{Z_i}{m}$
0	7	—
1	6	0.375
2	1	0.063
3	8	0.500
4	11	0.688
5	10	0.625
6	5	0.313
7	12	0.750
8	15	0.938
9	14	0.875
10	9	0.563
11	0	0.000
12	3	0.188
13	2	0.125
14	13	0.813
15	4	0.250
16	7	0.438
17	6	0.375
18	1	0.063
19	8	0.500

Chapter-8

* General approach to Generate Random ~~variable~~ variates:

2018-19 3(a)

* Inverse transform:

Suppose that we wish to generate a random variate X that is continuous and has distribution function F that is continuous and strictly increasing when $0 < F(x) < 1$.

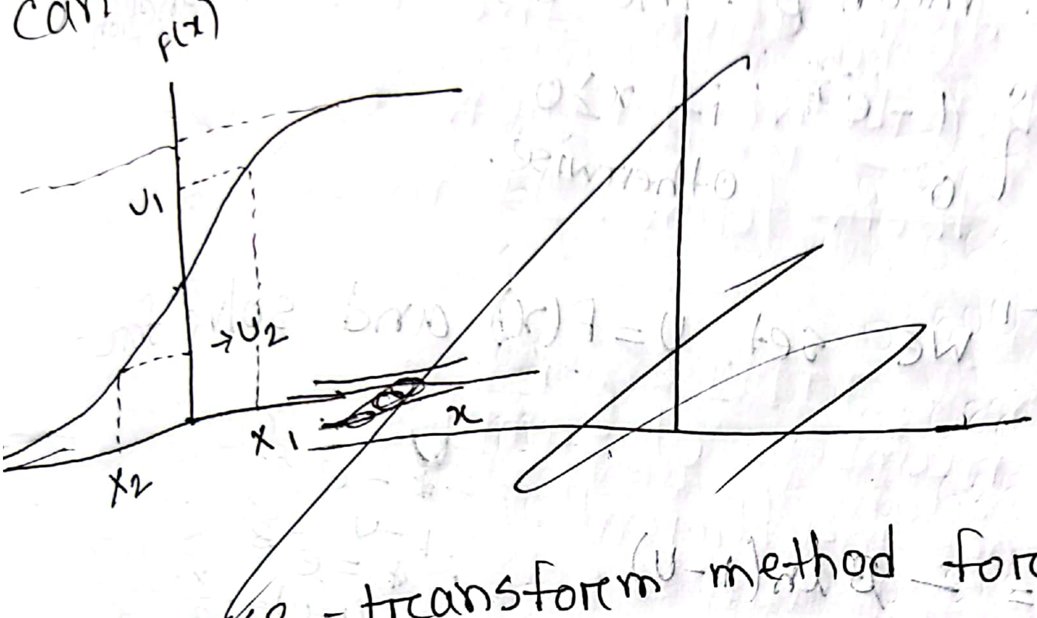
Let F^{-1} denote inverse of the function F. Then an algorithm for generating a random variate X having distribution function F is as follows:

$U = F(x)$
$F^{-1} U = X$

1. Generat ~~U~~ $U \sim U(0,1)$
2. Return $X = F^{-1}(U)$

$F^{-1}(U)$ will always be defined, since $0 \le U \le 1$ and the range of F is [0,1]

The illustration of this algorithm graphically is given here where the random variable corresponding to this distribution function can take on either positive or negative values.



ig: Inverse - transform method for continuous random variable

In the figure, number u_1 result in the positi-ve random variate x_1 and the random number u_2 result in the negative random variate x_2

or any real number x, $P(X \leq x) = F(x)$

ince, F is invertible, we have,

$$P(X \leq x) = P(F^{-1}(u) \leq x)$$

$$= P(U \leq F(x))$$

$$= F(x)$$

where, the last equality follows since $U \sim U(0,1)$ and $0 \le F(x) \le 1$.

2018-19 -3(a)

Example: 8.1 Let X have the exponential distribution with mean β. The distribution function is,

$$F(x) = \begin{cases} 1 - e^{-\frac{x}{\beta}} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Soln: To find F^{-1} we set $U = F(x)$ and solve for x to obtain,

$$U = 1 - e^{-\frac{x}{\beta}}$$

$$1 - U = e^{-\frac{x}{\beta}}$$

$$\ln(1-U) = -\frac{x}{\beta}$$

$$F^{-1}(U) = -\beta \ln(1-U)$$

Thus to generate the desired random variate, we first generate a $U \sim U(0,1)$ and then, let $X = -\beta \ln U$

when X is discrete, the distribution function is,

$$F(x) = P(X \le x) = \sum P(x_i)$$

where, $P(x_i)$ is the probability mass function

$$p(x_i) = P(X = x_i)$$

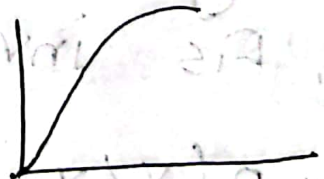


Fig: Inverse transform

The algorithm is then as follows:

1. Generate $U \sim U(0,1)$
2. Determine the smallest positive integer I such that $U \le F(x_i)$ and return $X = x_i$

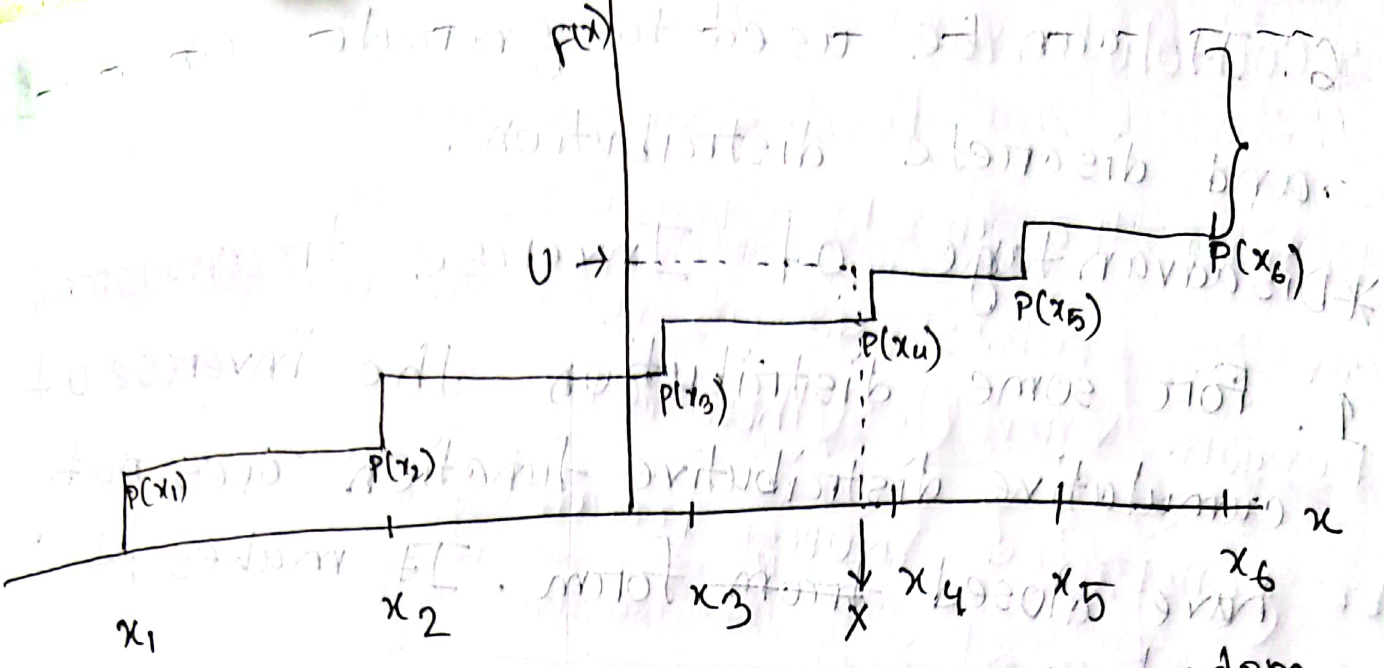


fig: Inverse transform method for discret random variate.

-Advantage of Inverse transform:

1. It can be applied to any distribution
2. The samples generated using the inverse transform method are theoretically exact.
3. The method is conceptually straightforward and simple.
4. It ensures that the generated random samples conform exactly to the desired distribution
5. It can be easily reproduced.

6. It can be used to generate continuous and discrete distribution.

*Disadvantage of Inverse transform:

1. For some distribution the inverse of cumulative distributive function does not have closed ~~from~~ form. It makes it complex.

2. For large number of samples the method becomes insufficient.

3. It requires expertise to solve the method.

*Composition:

The composition technique applies when the distribution function F from which we wish to generate can be expressed as a convex combination of other distribution function $F_1, F_2 \ldots$

Specifically, we assume that for all x, $F(x)$ is

$$F(x) = \sum_{j=1}^{\infty} P_j F_j(x) \text{ where, } P_j \geq 0 \cdot \sum P_j = 1$$

and each ~~of~~ F_j is distribution function.

The general algorithm:

1. Generate a positive random number J such that $P(J=j) = P_j$ for $j = 1, 2, 3 \ldots$
2. Return X with the distribution F_j

$$P(X \leq x) = \sum P(X \leq x \mid J = j) P(J = j) = \sum F_j(x) P_j = F(x)$$

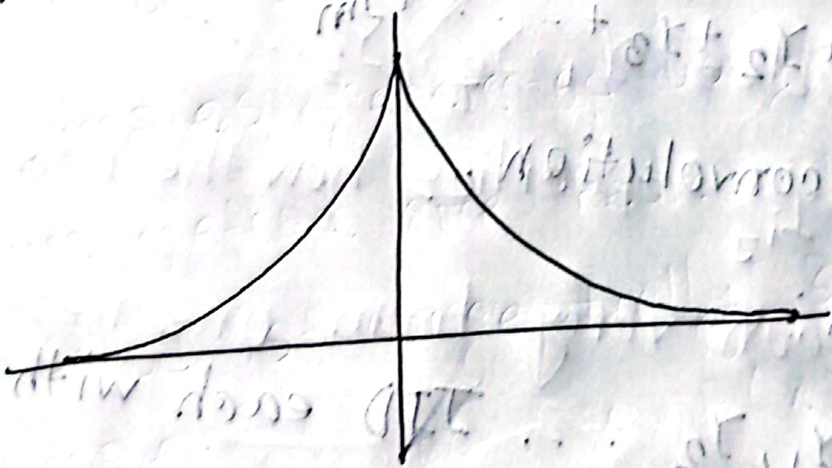


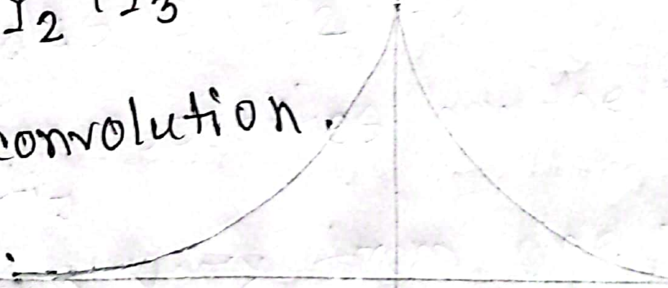
fig: Double exponential density.

* Convolution:

For several important distribution, the desired random variable X can be expressed as a sum of other random variables that are IID and can be generated more readily than direct generation of X,

We assume that IID random variables are $Y_1, Y_2, Y_3 \ldots$ such that $Y_1 + Y_2 + Y_3 + \ldots$ has the same distribution as X.

$$X = Y_1 + Y_2 + Y_3 + \ldots + Y_m$$



It is called convolution.

The algorithm:

1. Generate $Y_1, Y_2 \ldots$ IID each with distribution function G.
2. Return $X = Y_1 + Y_2 + Y_3 + \ldots + Y_m$.

Why output data analysis is important?

Soln:

Analysis of output data is important for many reasons:

1. Analysis of output data ensures that the model is accurately representing the real-world system.
2. By analyzing output data, one can verify the result.
3. Output data analysis provides potential outcomes, enabling informed decisions.
4. By examining output data, analysts can detect pattern anomalies.
5. Analyzing output data helps evaluating the efficiency, reliability and stability.
6. Output data analysis helps identifying bottlenecks
7. Output data analysis enables continuous refinement of the model.

* Transient and steady-state behavior of a stochastic process:

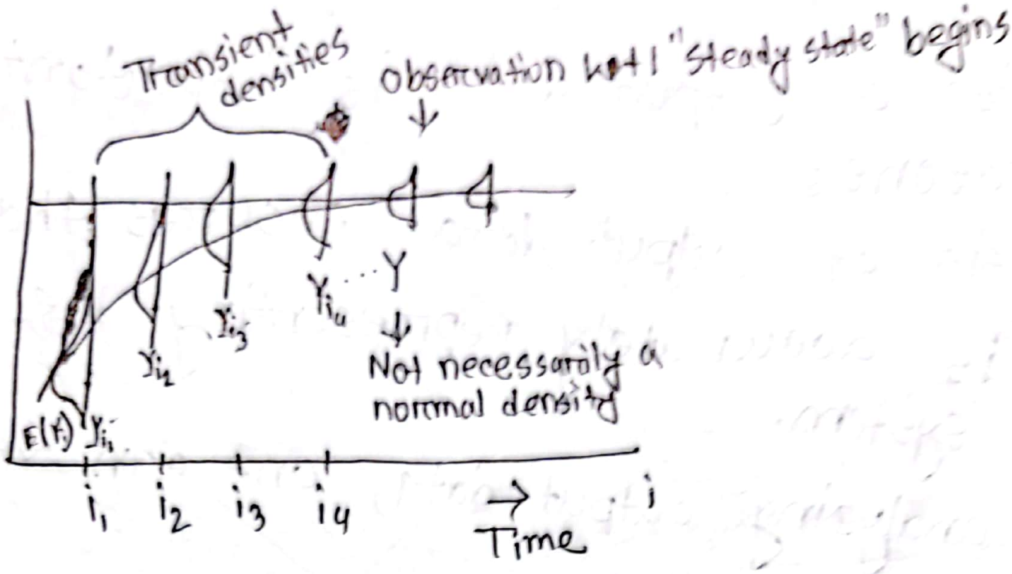


Fig: Transient and steady density functions for a particular stochastic process $Y_1, Y_2, \ldots$ and initial condition I.

Random variables $Y_1, Y_2, Y_3 \ldots$

The density $f_{Y_{i_j}}$ specifies how the random variable Y_{i_j} can vary from one replication to other.

For fixed y and I, the probabilities $F_1(y|I)$ $F_2(y|I)$ are just a sequence of numbers.

If $F_i(y|I) \rightarrow \cancel{F_i(y|I)\ \text{as } i}$ $F(y)$ as $i \to \infty$ for all y and I, $F(y)$ is called steady state distribution.

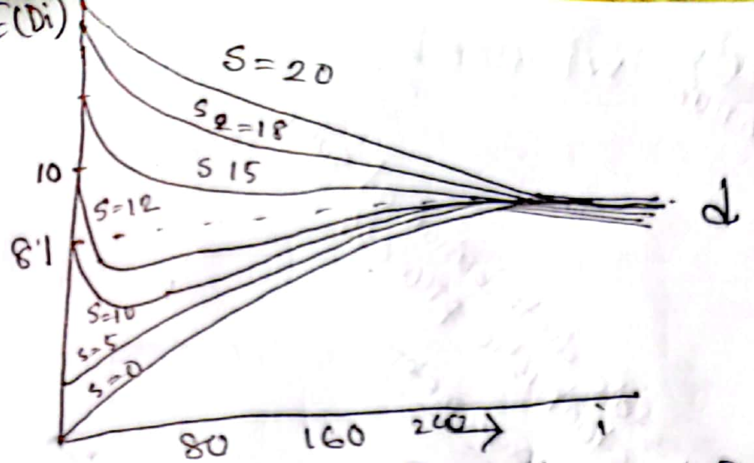


Fig: E (D_i) as a function of i and the number in system at time 0, s for the M/M/1 queue with $\rho = 0.9$

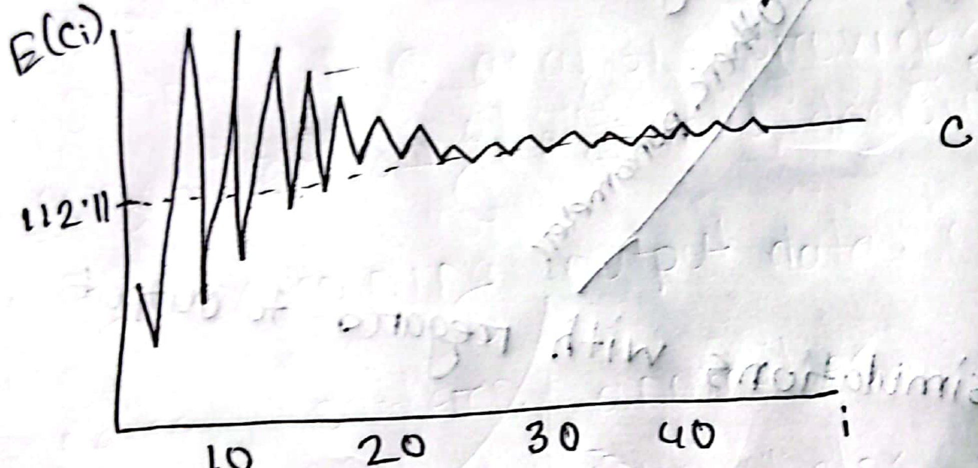


Fig: E(c_i) as a function of i for the (s, S) inventory system

* Types of simulation with regared to output Analysis:

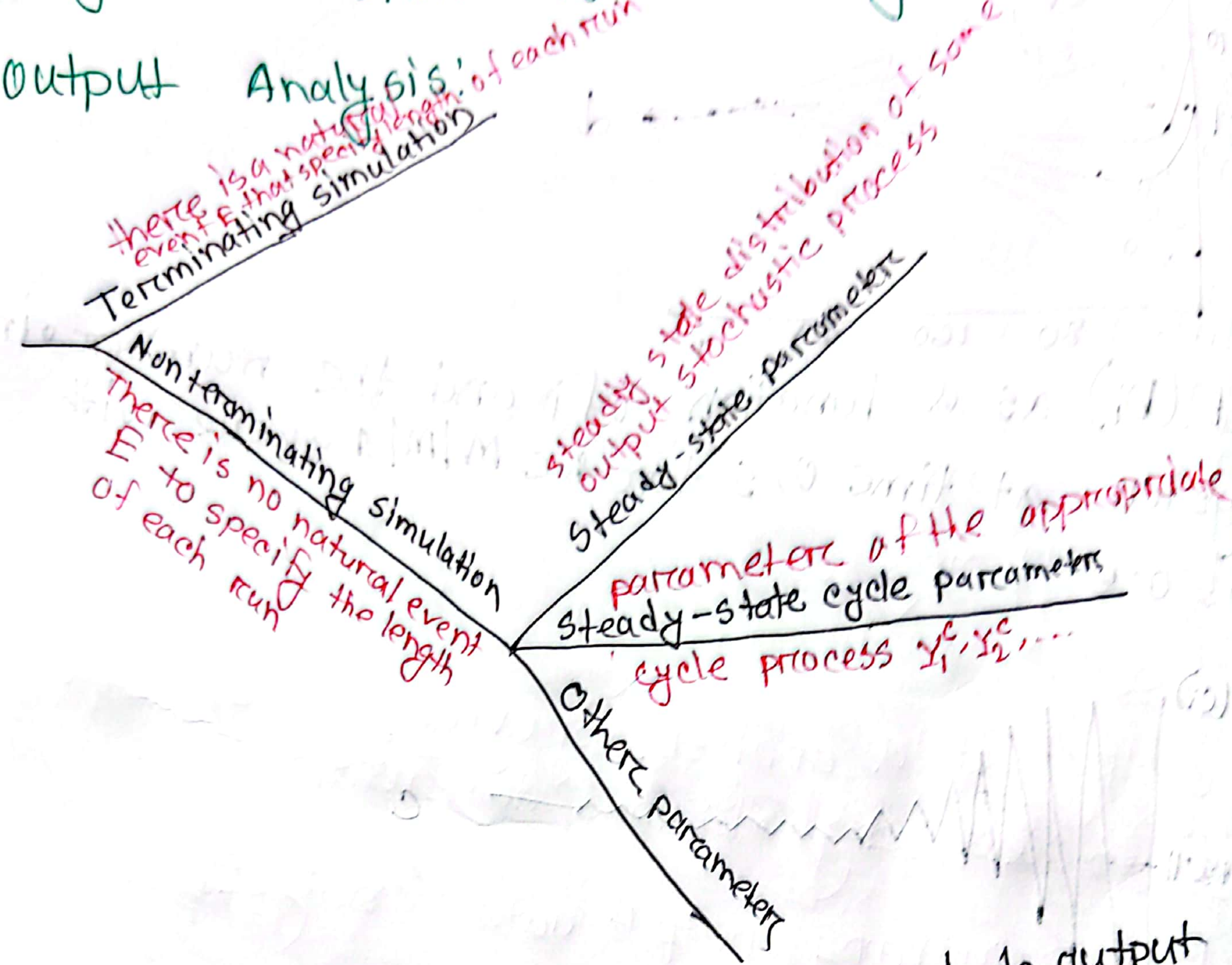


Fig: Types of simulations with regared to output analysis.

* Estimating Mean:

$$\text{Mean, } \mu = \bar{X}(n) \pm t_{n-1,1-\alpha/2}\sqrt{\frac{s^2(n)}{n}}$$

$X_1, \ldots X_n$ be the IID random variables.
IID → Independent Identically distributed

$s^2(n)$ = sample variance.

$0 < \alpha < 1$ → confidence level